Multiplying Bigger Numbers

Three-digit times two-digit: six pieces, or two rows.

243 × 36: six pieces

	200	40	3
30	6,000	1,200	90
6	1,200	240	18

Hundreds, tens and ones across the top; tens and ones down the side.

6,000 + 1,200 + 90 + 1,200 + 240 + 18 = 8,748

The same in two rows

$$\begin{array}{r} 243 \\ \times \quad 36 \\ \hline 1{,}458 \quad \times 6 \\ 7{,}290 \quad \times 30 \\ \hline 8{,}748 \end{array}$$

Row 1: 243 × 6 = 1,458 (the bottom pieces).
Row 2: 243 × 30 = 7,290 (the top pieces).
243 × 30 is 243 × 3 tens = 729 tens.
Add the two rows: 8,748.

Your turn!

312 × 23 = 7,176

For teachers

Goal: multiply three- and four-digit numbers by two-digit numbers (5.NBT.5). The two rows are the area model's pieces added in groups. Say “243 times 3 tens is 729 tens”, never “put down a zero”.

Multiplying Bigger Numbers: Practice

Split, multiply each piece, add.

Part 1 Hundreds times tens 3 hundreds × 4 tens = 12 thousands.

1) 300 × 40 = 12,000

2) 200 × 60 = 12,000

3) 500 × 70 = 35,000

4) 400 × 30 = 12,000

5) 700 × 20 = 14,000

6) 600 × 90 = 54,000

7) 800 × 50 = 40,000

8) 900 × 40 = 36,000

9) 200 × 80 = 16,000

10) 300 × 70 = 21,000

11) 500 × 50 = 25,000

12) 400 × 60 = 24,000

Multiplying Bigger Numbers: Practice (continued)

Part 2 Six pieces The parts are on the outside. Fill in each piece.

13) **234 × 15 = 3,510**

	200	30	4
10	2,000	300	40
5	1,000	150	20

2,000 + 300 + 40 + 1,000 + 150 + 20 = 3,510

14) **162 × 37 = 5,994**

	100	60	2
30	3,000	1,800	60
7	700	420	14

3,000 + 1,800 + 60 + 700 + 420 + 14 = 5,994

15) **425 × 23 = 9,775**

	400	20	5
20	8,000	400	100
3	1,200	60	15

8,000 + 400 + 100 + 1,200 + 60 + 15 = 9,775

Multiplying Bigger Numbers: Practice (continued)

Part 3 Split it yourself Write the parts on the outside, then fill in each piece.

16) **318 × 24 = 7,632**

	300	10	8
20	6,000	200	160
4	1,200	40	32

6,000 + 200 + 160 + 1,200 + 40 + 32 = 7,632

17) **256 × 43 = 11,008**

	200	50	6
40	8,000	2,000	240
3	600	150	18

8,000 + 2,000 + 240 + 600 + 150 + 18 = 11,008

18) **173 × 52 = 8,996**

	100	70	3
50	5,000	3,500	150
2	200	140	6

5,000 + 3,500 + 150 + 200 + 140 + 6 = 8,996

Multiplying Bigger Numbers: Practice (continued)

Part 4 Two rows Row 1: times the ones. Row 2: times the tens. Then add.

19)
$$\begin{array}{r} 214 \\ \times \quad 32 \\ \hline 428 \quad (\times 2) \\ 6{,}420 \quad (\times 30) \\ \hline 6{,}848 \end{array}$$

20)
$$\begin{array}{r} 365 \\ \times \quad 24 \\ \hline 1{,}460 \quad (\times 4) \\ 7{,}300 \quad (\times 20) \\ \hline 8{,}760 \end{array}$$

21)
$$\begin{array}{r} 128 \\ \times \quad 57 \\ \hline 896 \quad (\times 7) \\ 6{,}400 \quad (\times 50) \\ \hline 7{,}296 \end{array}$$

22)
$$\begin{array}{r} 432 \\ \times \quad 18 \\ \hline 3{,}456 \quad (\times 8) \\ 4{,}320 \quad (\times 10) \\ \hline 7{,}776 \end{array}$$

23)
$$\begin{array}{r} 251 \\ \times \quad 46 \\ \hline 1{,}506 \quad (\times 6) \\ 10{,}040 \quad (\times 40) \\ \hline 11{,}546 \end{array}$$

24)
$$\begin{array}{r} 617 \\ \times \quad 35 \\ \hline 3{,}085 \quad (\times 5) \\ 18{,}510 \quad (\times 30) \\ \hline 21{,}595 \end{array}$$

25)
$$\begin{array}{r} 183 \\ \times \quad 29 \\ \hline 1{,}647 \quad (\times 9) \\ 3{,}660 \quad (\times 20) \\ \hline 5{,}307 \end{array}$$

26)
$$\begin{array}{r} 526 \\ \times \quad 14 \\ \hline 2{,}104 \quad (\times 4) \\ 5{,}260 \quad (\times 10) \\ \hline 7{,}364 \end{array}$$

27)
$$\begin{array}{r} 342 \\ \times \quad 67 \\ \hline 2{,}394 \quad (\times 7) \\ 20{,}520 \quad (\times 60) \\ \hline 22{,}914 \end{array}$$

Part 5 Multiply No help this time.

28) $326 \times 21 = 6{,}846$

29) $147 \times 38 = 5{,}586$

Multiplying Bigger Numbers: Practice (continued)

Part 5 Multiply (continued)

30) 512 × 46 = 23,552

31) 238 × 17 = 4,046

32) 403 × 25 = 10,075

33) 671 × 34 = 22,814

34) 295 × 42 = 12,390

35) 184 × 63 = 11,592

36) 736 × 15 = 11,040

37) 459 × 27 = 12,393

38) 128 × 74 = 9,472

39) 387 × 56 = 21,672

40) 604 × 39 = 23,556

41) 215 × 88 = 18,920

Multiplying Bigger Numbers: Practice (continued)

Part 5 Multiply (continued)

42) 548 × 23 = 12,604

43) 963 × 12 = 11,556

44) 372 × 45 = 16,740

45) 819 × 26 = 21,294

46) 256 × 91 = 23,296

47) 437 × 68 = 29,716

Part 6 Four digits times one digit No help this time.

48) 2,134 × 3 = 6,402

49) 4,206 × 7 = 29,442

50) 1,573 × 4 = 6,292

51) 3,058 × 6 = 18,348

52) 6,125 × 8 = 49,000

53) 2,947 × 5 = 14,735

Multiplying Bigger Numbers: Practice (continued)

Part 6 Four digits times one digit (continued)

54) 5,312 × 9 = 47,808

55) 1,806 × 4 = 7,224

56) 7,243 × 3 = 21,729

57) 3,615 × 6 = 21,690

Part 7 Multiply

58) 672 × 35 = 23,520

59) 763 × 43 = 32,809

60) 728 × 41 = 29,848

61) 163 × 54 = 8,802

62) 353 × 47 = 16,591

63) 182 × 39 = 7,098

64) 672 × 59 = 39,648

65) 236 × 54 = 12,744

Multiplying Bigger Numbers: Practice (continued)

Part 7 Multiply (continued)

66) 526 × 16 = 8,416

67) 162 × 15 = 2,430

68) 621 × 38 = 23,598

69) 113 × 29 = 3,277

70) 873 × 19 = 16,587

71) 113 × 22 = 2,486

72) 194 × 12 = 2,328

73) 664 × 59 = 39,176

74) 153 × 33 = 5,049

75) 542 × 49 = 26,558

76) 371 × 26 = 9,646

77) 736 × 48 = 35,328

Multiplying Bigger Numbers: Practice (continued)

Part 7 Multiply (continued)

78) 358 × 17 = 6,086

79) 848 × 59 = 50,032

80) 237 × 38 = 9,006

81) 275 × 42 = 11,550

82) 344 × 58 = 19,952

83) 355 × 48 = 17,040

84) 148 × 17 = 2,516

85) 464 × 12 = 5,568

86) 497 × 24 = 11,928

87) 527 × 55 = 28,985

88) 459 × 14 = 6,426

89) 604 × 24 = 14,496

Multiplying Bigger Numbers: Practice (continued)

Part 7 Multiply (continued)

90) 182 × 36 = 6,552

91) 883 × 43 = 37,969

92) 121 × 13 = 1,573

93) 165 × 47 = 7,755

Part 8 Remember? Problems from earlier packets.

94) 3.6 + 2.75 = 6.35

95) 8.2 − 4.67 = 3.53

96) 4.28 to the nearest tenth = 4.3

97) 0.45 > 0.405

98) in 3.142 the 4 is worth 0.04

99) 2.7 × 1,000 = 2,700

Multiplying Bigger Numbers: Practice (continued)

Part 9 Word problems Split, multiply, add.

100) A school bus holds 48 kids. How many kids can 125 buses hold?

Number sentence **125 × 48 = 6,000**

Answer **6,000** kids

Draw or show your work

101) A farm packs 236 boxes of eggs. Each box has 12 eggs. How many eggs is that?

Number sentence **236 × 12 = 2,832**

Answer **2,832** eggs

Draw or show your work

102) A plane flies 465 miles each day for 21 days. How far does it fly?

Number sentence **465 × 21 = 9,765**

Answer **9,765** miles

Draw or show your work

103) A box of paper has 1,250 sheets. How many sheets are in 8 boxes?

Number sentence **1,250 × 8 = 10,000**

Answer **10,000** sheets

Draw or show your work

104) Challenge: A theater has 34 rows of 28 seats. 875 seats are full. How many are empty?

Number sentence **34 × 28 = 952, 952 − 875 = 77**

Answer **77** seats

Draw or show your work

Check-In: Multiplying Bigger Numbers

Show what you know. No help this time.

Mastery: 8 of 10 (80%)

Part 1 Hundreds times tens

1) 300 × 90 = 27,000

2) 800 × 90 = 72,000

3) 700 × 70 = 49,000

Part 2 Two rows

4)

418 × 23

1,254 × 3

8,360 × 20

9,614

5)

152 × 47

1,064 × 7

6,080 × 40

7,144

6)

324 × 26

1,944 × 6

6,480 × 20

8,424

Part 3 Multiply

7) 382 × 19 = 7,258

8) 527 × 34 = 17,918

9) 243 × 17 = 4,131

10) 275 × 36 = 9,900